AOC 2026 Singapore

北京大学 PEKING UNIVERSITY

Disentangling climate signals from survey effort in new bird distribution records across China

Birds or birders? A discrete-time hazard model of first records, 2002–2024

Chenchen Ding[1], Zhengwang Zhang[2], Zhigang Jiang[3], Zhiheng Wang[1]

[1] Peking University [2] Beijing Normal University [3] Institute of Zoology, Chinese Academy of Sciences

3rd Asian Ornithological Conference · National University of Singapore · 14–17 November 2026

Background

Global change is moving species, but we see it only through incomplete records

1 Global change threatens about one million species;
Species are moving poleward and upslope,
16.9 km per decade on land

IPBES 2019; Chen et al. 2011; Pecl et al. 2017

2 Tracking these moves needs distribution data;
Those data are incomplete, and unevenly so
across taxa, regions and time

Lomolino 2004; Hortal et al. 2015; Herkt et al. 2017

3 New distribution records are where a move first shows;
But a record shows that a bird was found,
not why it was there

Hughes et al. 2021; Lawlor et al. 2024

Biases accumulate at four stages

1 **Field survey**
uneven expertise, funding and access

2 **Detection**
detectability, effort and survey method

3 **Documentation**
local journals, other languages, old taxonomy

4 **Data sharing**
slow, partial entry into GBIF, IUCN, BirdLife

Boakes et al. 2010; Meyer et al. 2015; Oliver et al. 2021

Biases in how we look can be mistaken for patterns in nature

Background

China is a key region for bird distributions, but its new records are scattered

Zoogeographic realms
2
Palearctic and Oriental meet

Biodiversity hotspots
4
global hotspots within China

Migratory crossroads
Flyways
several major flyways cross it

Bird species
1,505
about 14% of the world's birds

Zhang 2011; Mittermeier et al. 2011; Zheng 2023

Monitoring has expanded fast
National networks such as Sino BON

Citizen science and camera traps

Sound recording and DNA identification

Feng et al. 2019; Huang et al. 2021; Mi et al. 2021

But the evidence is scattered
Chinese-language papers and local reports

Mixed taxonomy, few coordinates

Largely absent from global databases

Xing et al. 2024; Chen et al. 2025

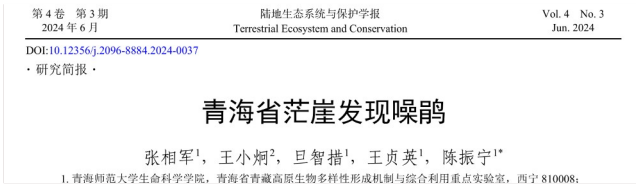
Gap: no standardized, open dataset of provincial new bird records in China
So we built one: CBNR, the China Bird New Record dataset

Background

Each new record is a short paper, backed by a photograph



Qinghai · found 2023, published 2024



第 4 卷 第 3 期 2024 年 6 月 | 陆地生态系统与保护学报 Terrestrial Ecosystem and Conservation | Vol. 4 No. 3 Jun. 2024

DOI:10.12356/j.2096-8884.2024-0037

· 研究简报 ·

青海省茫崖发现噪鹃

张相军[1]，王小炯[2]，旦智措[1]，王贞英[1]，陈振宁[1*]

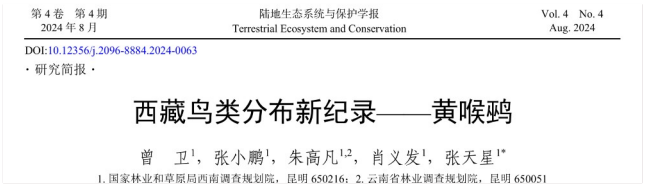
1. 青海师范大学生命科学学院，青海省青藏高原生物多样性形成机制与综合利用重点实验室，西宁 810008；

Western Koel

Eudynamys scolopaceus



Xizang · found 2023, published 2024



第 4 卷 第 4 期 2024 年 8 月 | 陆地生态系统与保护学报 Terrestrial Ecosystem and Conservation | Vol. 4 No. 4 Aug. 2024

DOI:10.12356/j.2096-8884.2024-0063

· 研究简报 ·

西藏鸟类分布新纪录——黄喉鹀

曾 卫[1]，张小鹏[1]，朱高凡[1,2]，肖义发[1]，张天星[1*]

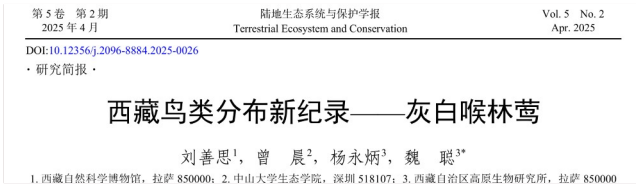
1. 国家林业和草原局西南调查规划院，昆明 650216；2. 云南省林业调查规划院，昆明 650051

Yellow-throated Bunting

Emberiza elegans



Xizang · found 2024, published 2025



第 5 卷 第 2 期 2025 年 4 月 | 陆地生态系统与保护学报 Terrestrial Ecosystem and Conservation | Vol. 5 No. 2 Apr. 2025

DOI:10.12356/j.2096-8884.2025-0026

· 研究简报 ·

西藏鸟类分布新纪录——灰白喉林莺

刘善思[1]，曾 晨[2]，杨永炳[3]，魏 聪[3*]

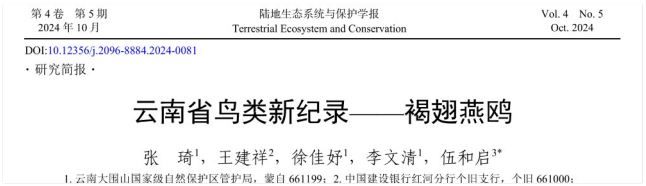
1. 西藏自然科学博物馆，拉萨 850000；2. 中山大学生态学院，深圳 518107；3. 西藏自治区高原生物研究所，拉萨 850000

Common Whitethroat

Curruca communis



Yunnan · found 2024 and published



第 4 卷 第 5 期 2024 年 10 月 | 陆地生态系统与保护学报 Terrestrial Ecosystem and Conservation | Vol. 4 No. 5 Oct. 2024

DOI:10.12356/j.2096-8884.2024-0081

· 研究简报 ·

云南省鸟类新纪录——褐翅燕鸥

张 琦[1]，王建祥[2]，徐佳妤[1]，李文清[1]，伍和启[3*]

1. 云南大围山国家级自然保护区管护局，蒙自 661199；2. 中国建设银行红河分行个旧支行，个旧 661000；

Bridled Tern

Onychoprion anaethetus



Hunan · found 2024, published 2025



野生动物学报 Chinese Journal of Wildlife 2025,46(3):692－697

Chinese Journal of Wildlife

野生动物学报

http://ysdw.nefu.edu.cn

数据资源 RESOURCES

湖南省七种鸟类新纪录

Asian Openbill

Anastomus oscitans



Henan · found 2022, published 2024



野生动物学报 Chinese Journal of Wildlife 2024,45(3):675－677

Chinese Journal of Wildlife

野生动物学报

http://ysdw.nefu.edu.cn

河南省鸟类分布新纪录——褐马鸡

Brown Eared-Pheasant

Crossoptilon mantchuricum

Open-access sources: Terrestrial Ecosystem and Conservation 2024–2025; Chinese Journal of Wildlife 2024–2025

Background

A new record is the last link in a chain of three events

Provincial new record: the first verified occurrence of a known species in a province where it was not reported before

1 The bird is there



Climate, land use or vagrancy can move a species in

Ecological process

2 Someone looks



More surveys, better access and new methods find what was missed

Observation process

3 The find is published



The detection is written up and dated in a paper or report

Documentation

New record

Only this link is ecological

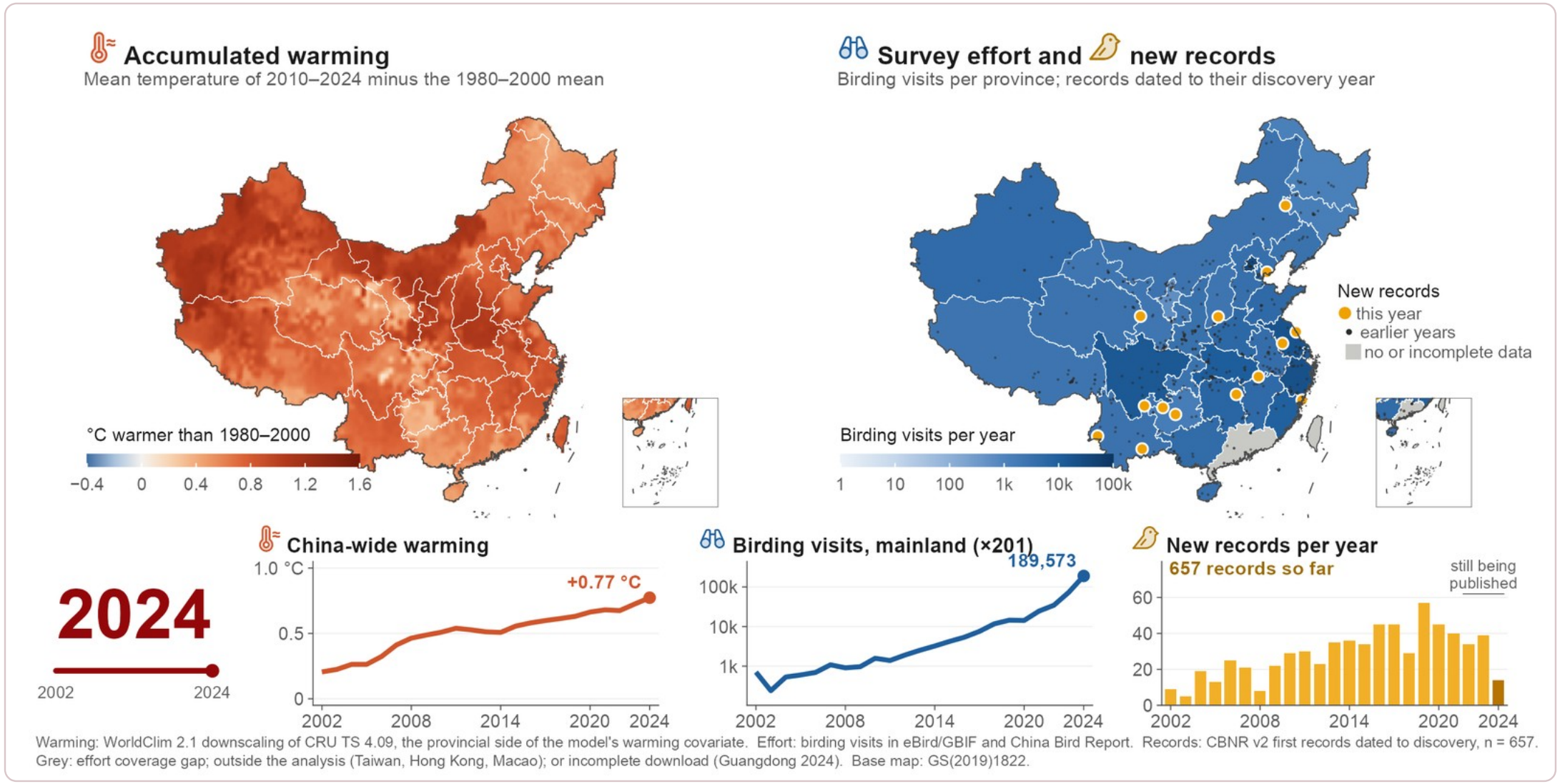
These two leave the same trace in the data

- [ ] Two opposite errors: reading every record as a range shift, or dismissing all as sampling noise

Chen et al. 2011; Pecl et al. 2017; Lawlor et al. 2024; Lahoz-Monfort & Magrath 2021; Sanczuk et al. 2026

Background

After 2002, warming, birding and new records all rose together



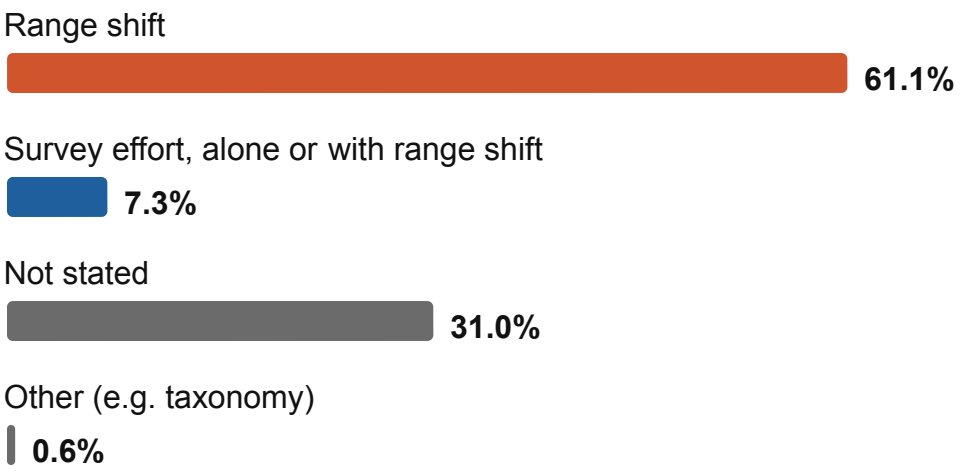
Warming: WorldClim 2.1 downscaling of CRU TS 4.09, the provincial side of the model's warming covariate. Effort: birding visits in eBird/GBIF and China Bird Report. Records: CBNR v2 first records dated to discovery, n = 657. Grey: effort coverage gap; outside the analysis (Taiwan, Hong Kong, Macao); or incomplete download (Guangdong 2024). Base map: GS(2019)1822.

Research gaps

Analyses of new records have not separated ecology from observation

How source papers explain their records

CBNR v2, n = 1,029 provincial first records



Birding visits rose about 200-fold, 2002–04 to 2022–24, yet effort is rarely given as the reason

1 GAP

Drivers studied apart

Range shifts or sampling bias, usually one at a time

Warming and effort are rarely modelled together; their interaction is almost never tested

2 GAP

Heterogeneity unexplored

Responses differ among species, places and years

Do traits such as migration modulate both drivers?
Do records lean poleward, net of geometry?

3 GAP

Scattered data, mismatched scales

Records are local, scattered and dated to publication

No national, discovery-dated dataset links each record to annual climate and effort

Quantifying both drivers, their interaction and its heterogeneity nationally fills this gap

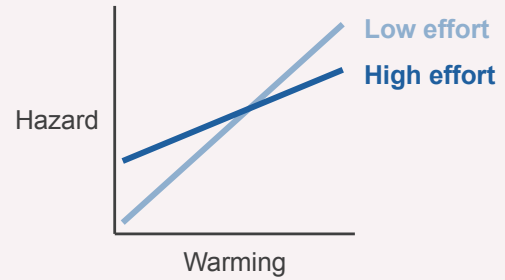
Boakes et al. 2010; Pacifici et al. 2017; Lahoz-Monfort & Magrath 2021; Xing et al. 2024

Questions

Three questions, each with a testable hypothesis

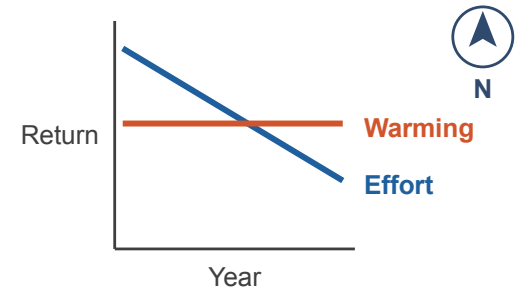
Q1 How do warming and survey effort jointly drive new records?

H1 Both raise the annual hazard of a first record, warming beyond its regional-climate null; their effects are comparable and interact negatively: where effort is high, the species that warming favours are already on the list



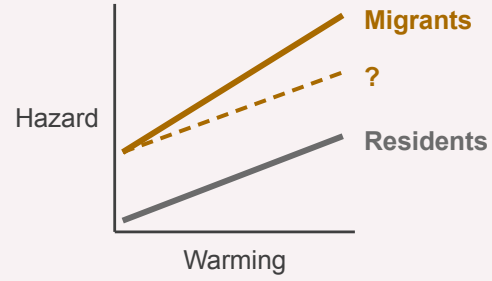
Q2 Do their effects change through time and across space?

H2 Province-year survey pulses dominate the variation; the return on effort declines as gaps fill, the return on warming does not; net of geometry and effort, records lean poleward



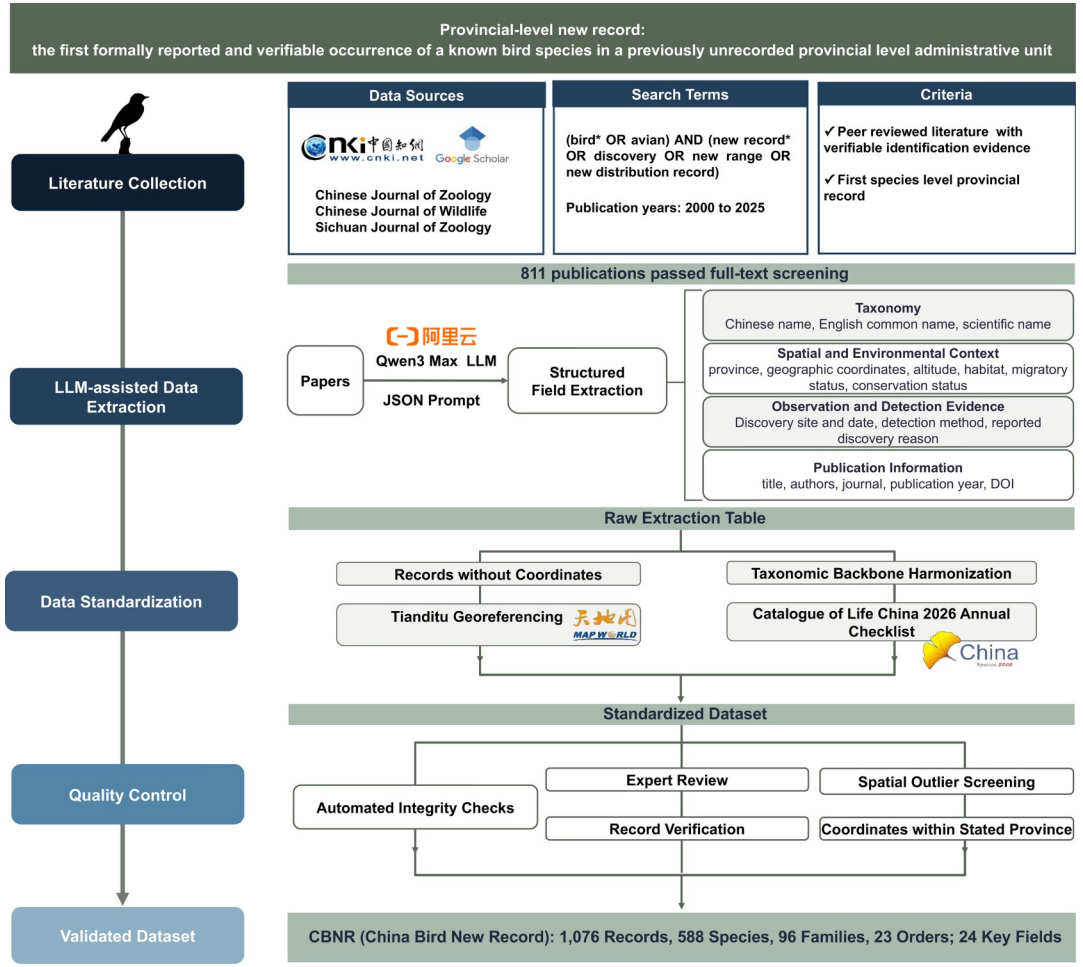
Q3 Which species gain records, and do traits modulate the two drivers?

H3 Mobile, wide-ranging species, such as partial migrants, gain more records; migratory strategy may change how strongly warming and effort act (for example, a weaker warming effect in long-distance migrants)



Data

CBNR turns 25 years of scattered literature into one auditable dataset



CBNR Fig. 1 Technical workflow

1 Literature
CNKI, Google Scholar and specialist journals, 2000–2025

2 LLM-assisted extraction
Qwen3-Max with a fixed JSON schema

3 Standardization
Tianditu coordinates; Catalogue of Life China 2026

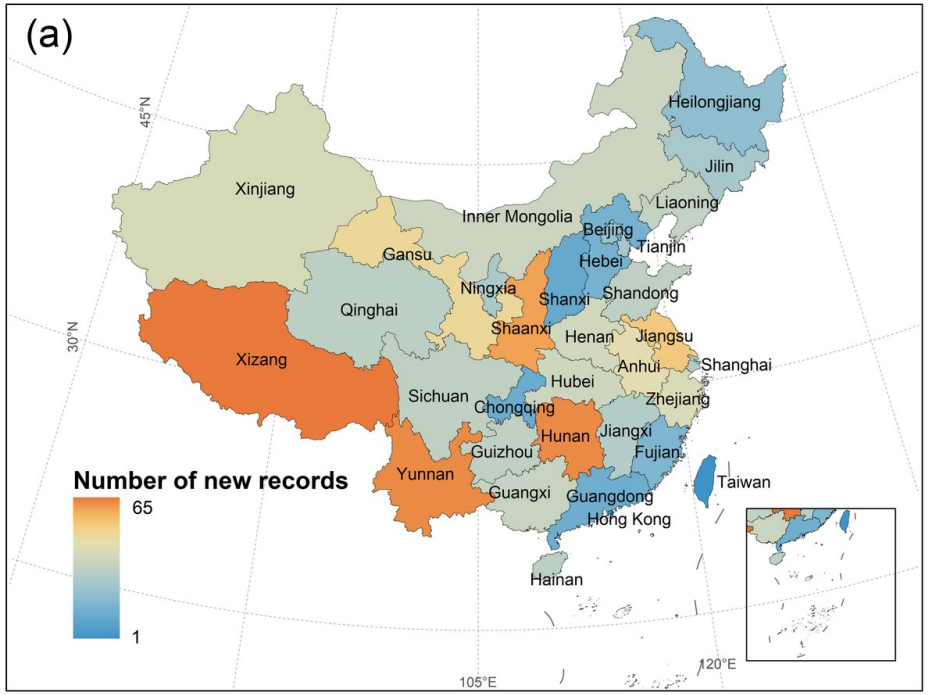
4 Quality control
Integrity checks, expert review, spatial outliers

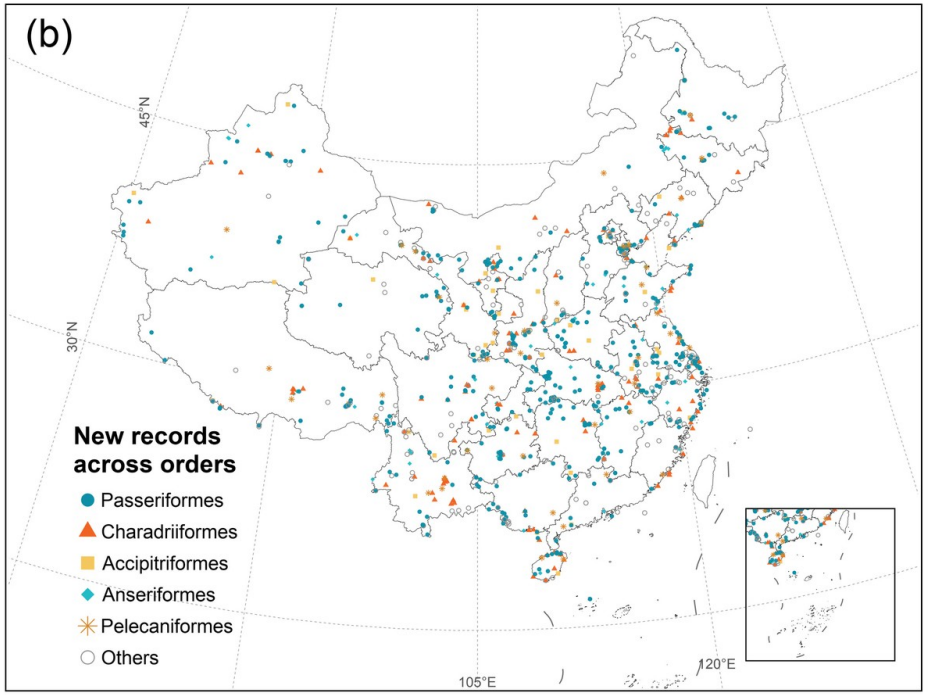
5 Release
1,076 records · 588 species · 23 orders

797 papers · 33 provinces · 24 key fields
Every curation step logged

Data

New records come from every province, most from the south-west and centre





CBNR Fig. 2a, b Records per province; localities by order

Xizang
65
most records of any province

Hunan · Yunnan
63 · 63
the next highest

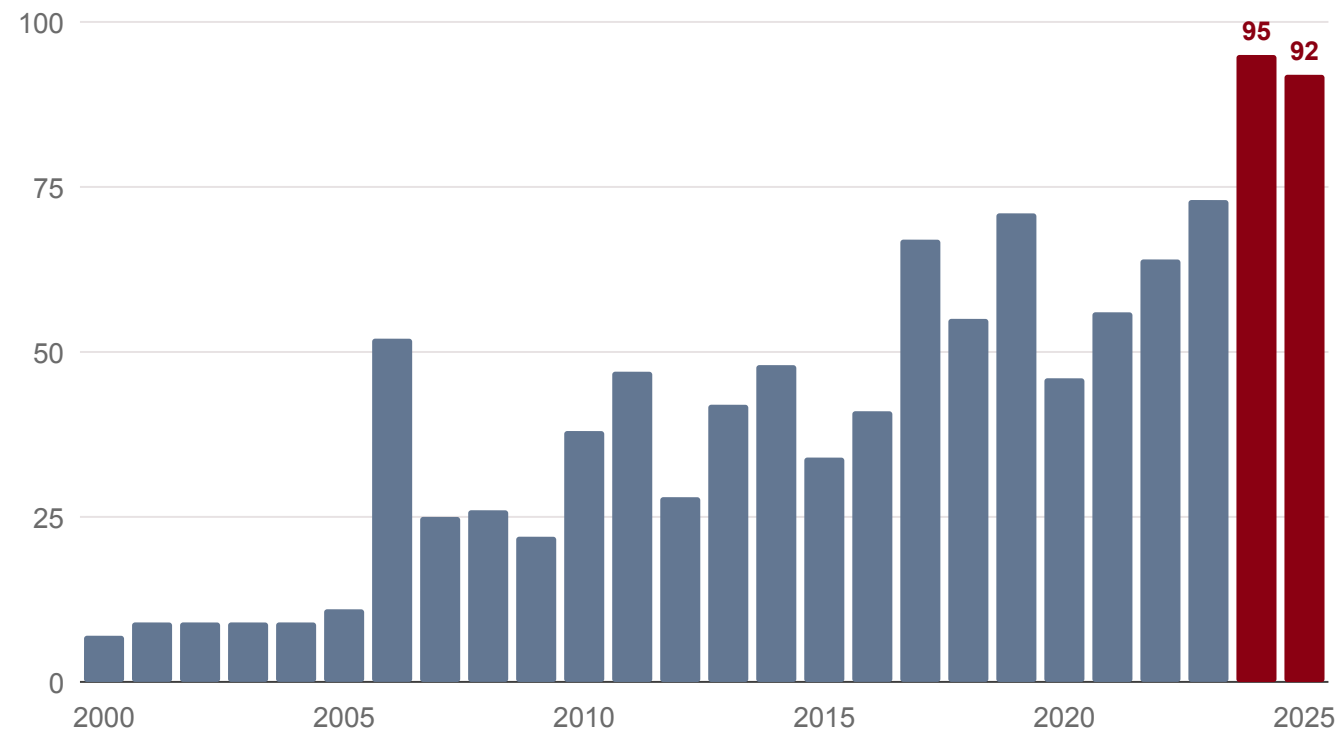
Passeriformes
48.4%
521 records, 318 species

Field photographs
79%
the commonest evidence

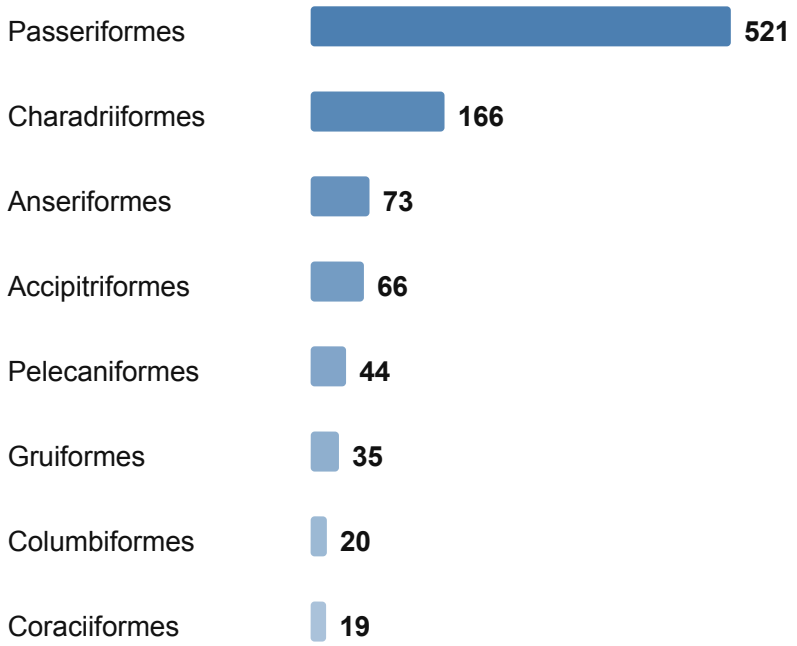
Data

Records rose after 2010, led by songbirds and shorebirds

New records per publication year



Records by order



Coverage of each order's Chinese species: 32–66%

Records rose after 2010 and are concentrated in songbirds and shorebirds

Data

Wetland and soaring orders are best covered; phylogenetic signal is weak



CBNR Fig. 3 New records on the bird phylogeny; bubbles: share of each order's species with a record

Fritz–Purvis D = 0.80: weak phylogenetic structure; no clustering (P = 0.87)

Approach

From records to a hazard model, in four steps

1 Records

- **CBNR v2** — 1,029 provincial first records
- **Dated to discovery** — 657 records in 2002–2024
- **Current release, v1.1** — 1,076 records · 588 species

2 Risk set

- **Species × province × year** — 175,640 cells, 2002–2024
- **Reachable provinces only** — ≥ 50 suitable SDM cells
- **Exit after first record** — each pair counted once

3 Covariates

- **Warming** — vs species range, 15 yr
- **Survey effort** — birding visits
- **Variability** — annual anomaly
- **Completeness** — offset, recent years

4 Analysis

- **Hazard GLMM** — cloglog link
- **Robustness** — nulls · 12 specs
- **Through time** — time-varying returns
- **Projections** — support checked

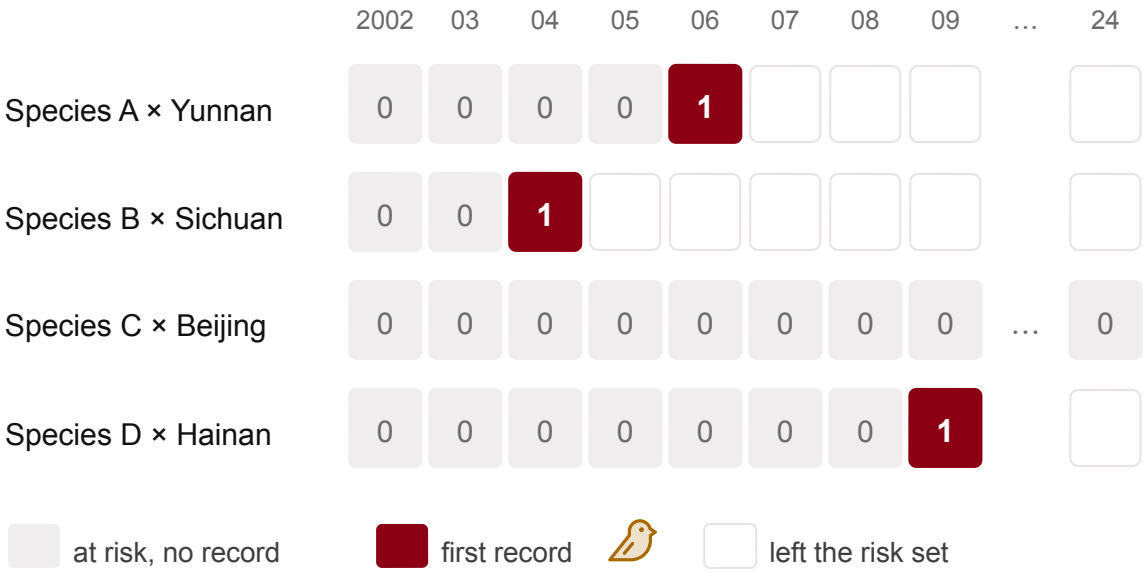
Outputs: contributions · interaction · observation process · returns through time · futures

Approach

A discrete-time hazard model asks: is this the year of the first record?

The risk set

Each pair is at risk every year until its first record



175,640 cells · 649 first records

392 species · 31 provinces · 2002–2024

first record ~ warming × effort + variability
+ offset(log completeness)
+ (1 | species) + (1 | province) + (1 | province × year)

Binomial, cloglog link: exp(β) is a hazard ratio per SD

Accumulated warming

Province minus species-range anomaly, 15-yr; 1 SD = 0.18 °C

Survey effort

Provincial birding visits; 1 SD = 7.9-fold more visits

Province × year effect

Survey campaigns, birding festivals, new reserves

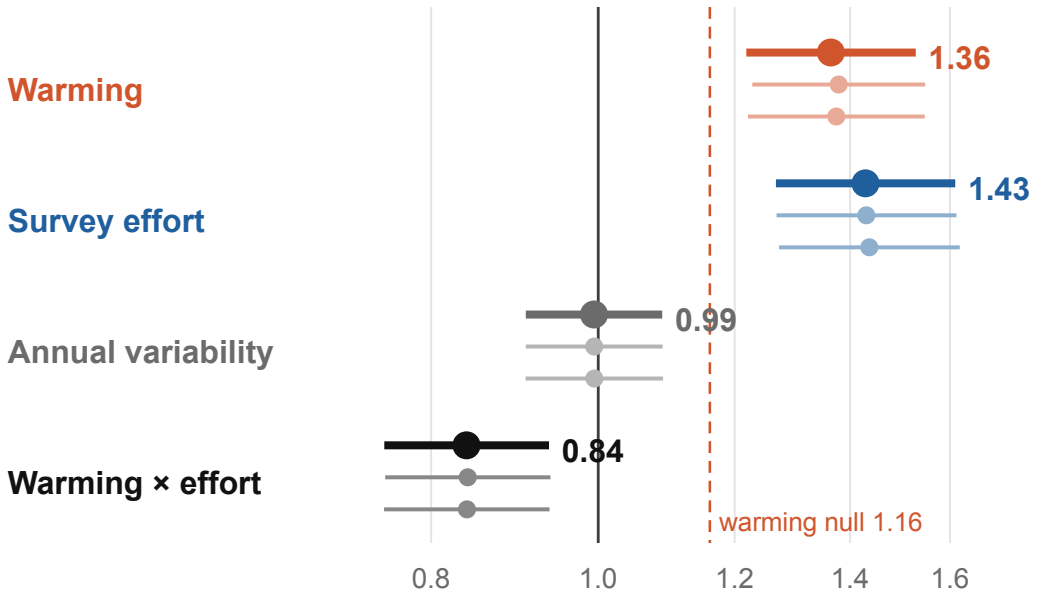
Dated to discovery

Offset corrects the latest, incompletely published years

◆ Finding 1

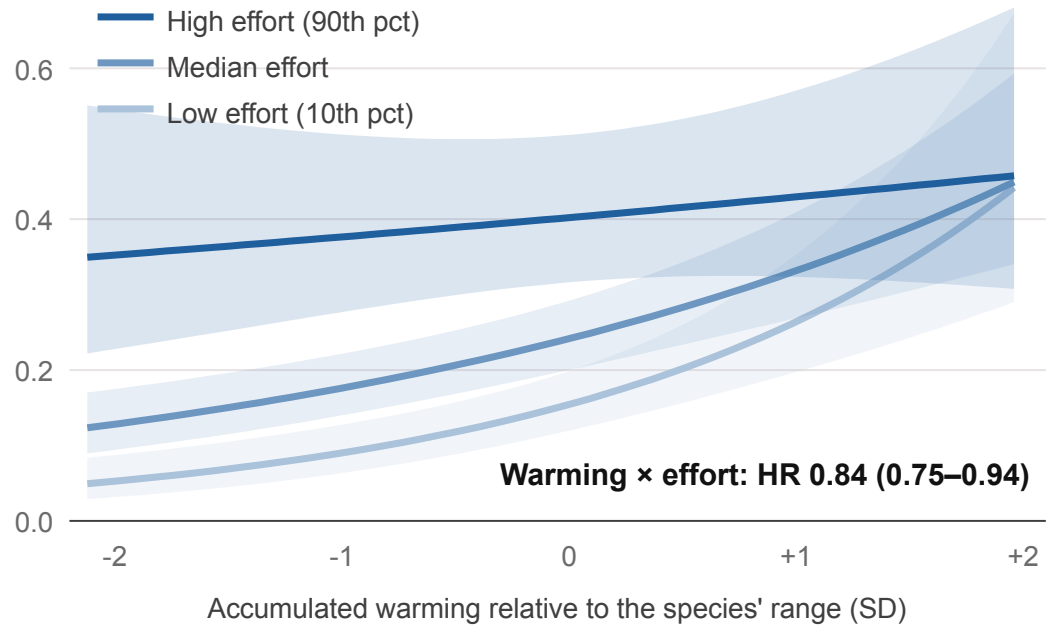
Warming and survey effort contribute comparably, and they interact

Hazard ratio per SD (95% CI)



Thick: main analysis (≥ 50 suitable SDM cells); thin: 100 and 200 cells; log scale

Annual hazard of a first record (%)



Accumulated warming
1.36
HR per SD (1.22–1.53)

Survey effort
1.43
HR per SD (1.27–1.61)

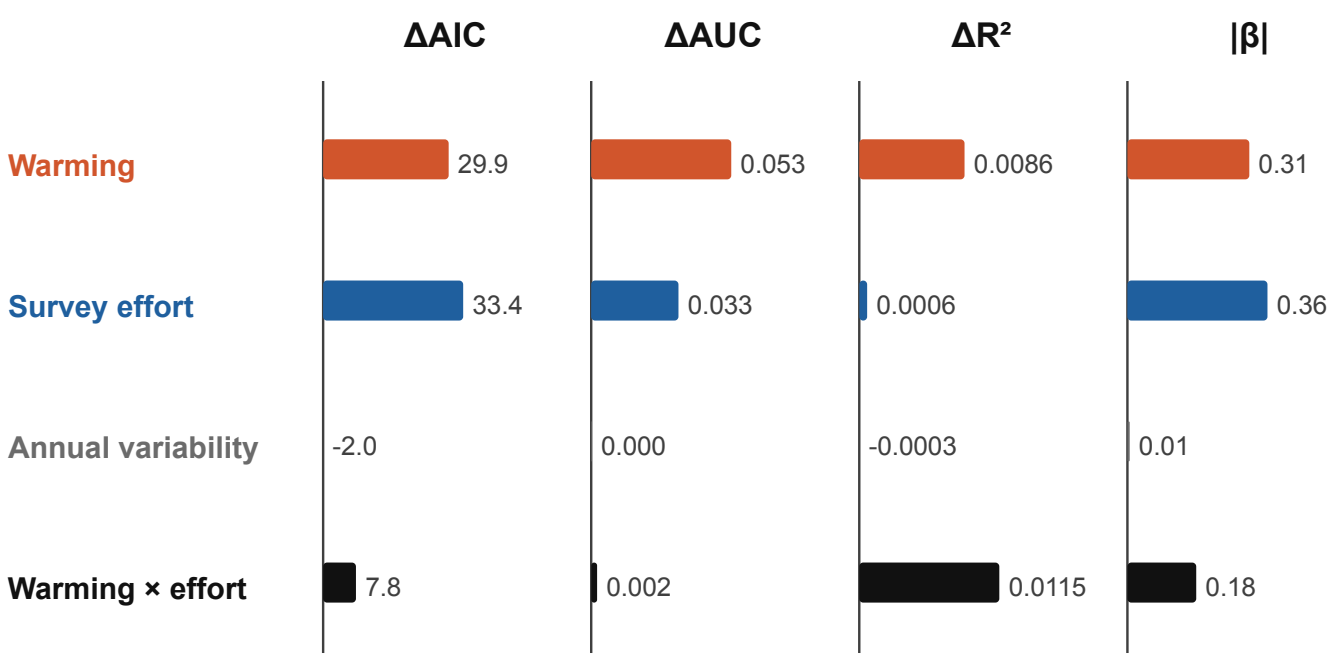
Warming × effort
0.84
steepest where few look

Annual variability
0.99
no effect (0.91–1.09)

Finding 1

The balance holds on average across criteria and twelve specifications

Loss in fit when the term is dropped



Each column scaled to its own maximum; |β|: absolute standardised coefficient

Warming and effort rank equally on average across the four criteria

Cost of dropping a term

ΔAIC: effort 33.4, warming 29.9
ΔAUC: warming 0.053, effort 0.033
|β|: effort 0.36, warming 0.31

Twelve specifications

SDM thresholds 50, 100, 200
4 effort proxies × 3 missing-data rules
Effort 1.25–1.55; warming 1.33–1.39

Design-preserving nulls

Warming null 1.16: increment ×1.18
Shuffled effort timing: 0.98
Province bootstrap: lower bounds > 1

Finding 2

Survey pulses are the largest source of variation in the hazard

Observation shocks vary by province and year

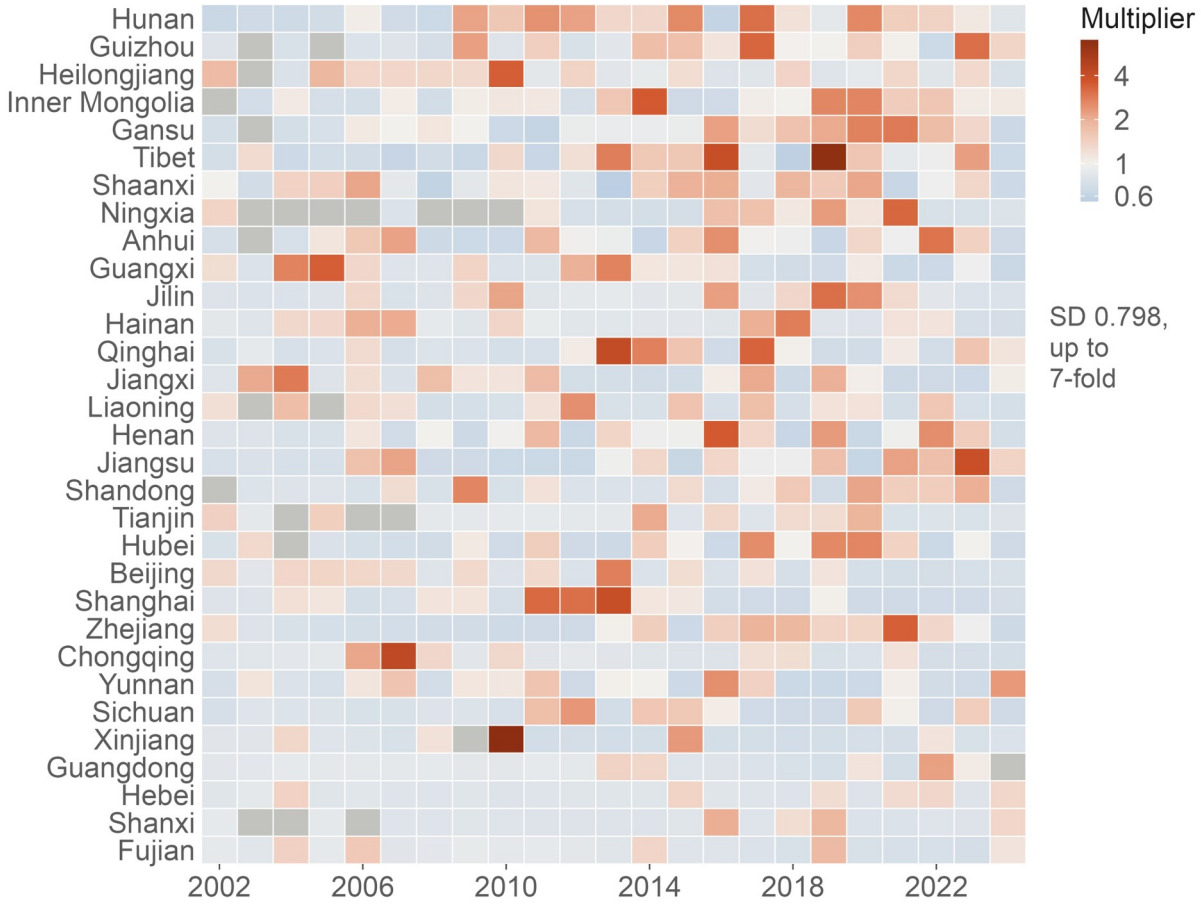
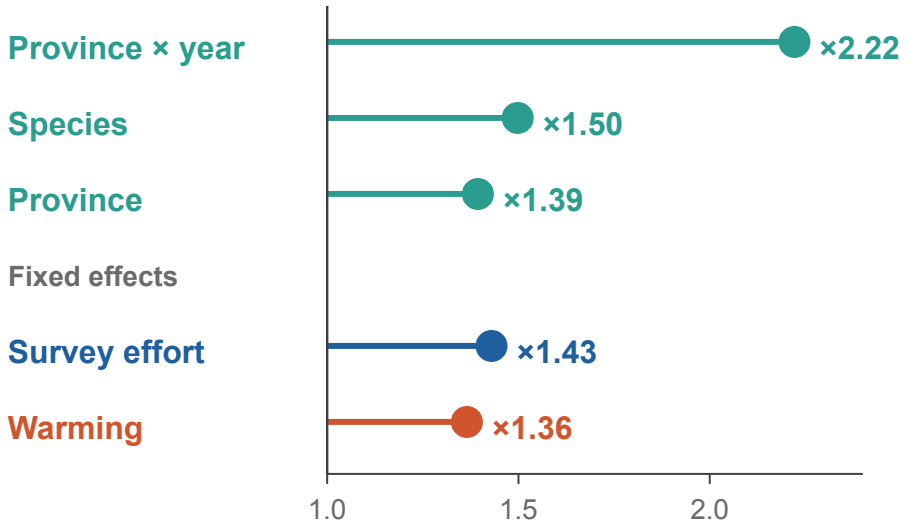


Fig. 5c Province × year multipliers

Hazard multiplier per SD



Conditional AUC

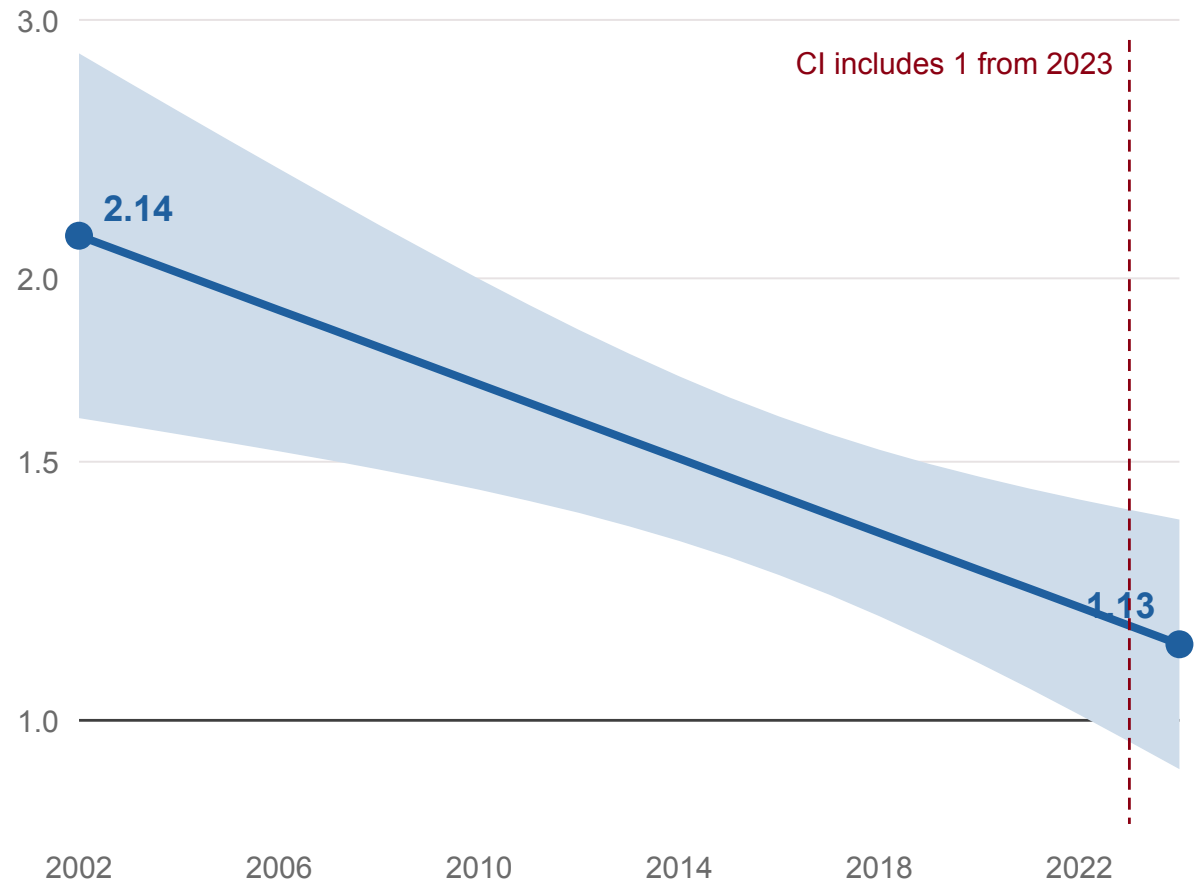
0.72 → 0.85

after adding the province × year layer

Finding 3

The return on effort is falling; the return on warming is not

Return on survey effort: HR per SD (95% CI)



Return on effort

2.14 → 1.13

HR per SD, 2002 → 2024; P = 0.0021

Return on warming

P = 1.00

warming × time: no trend; HR 1.37

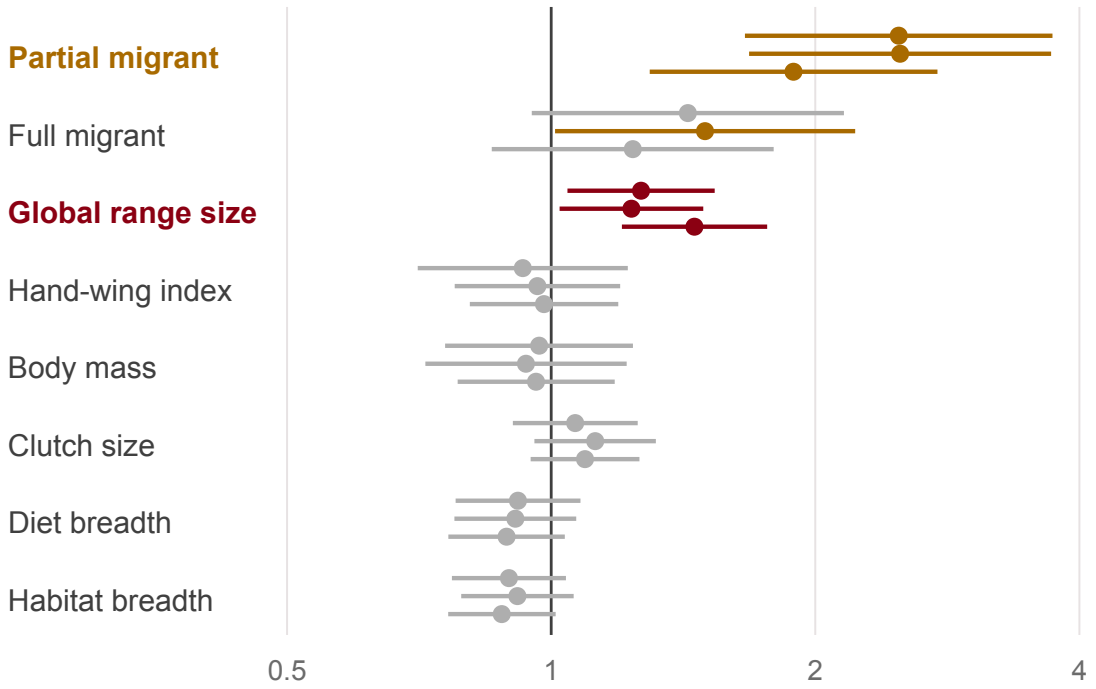
A stock and a flow

Effort mines a finite stock of undocumented occurrences; warming keeps adding new candidates

Traits

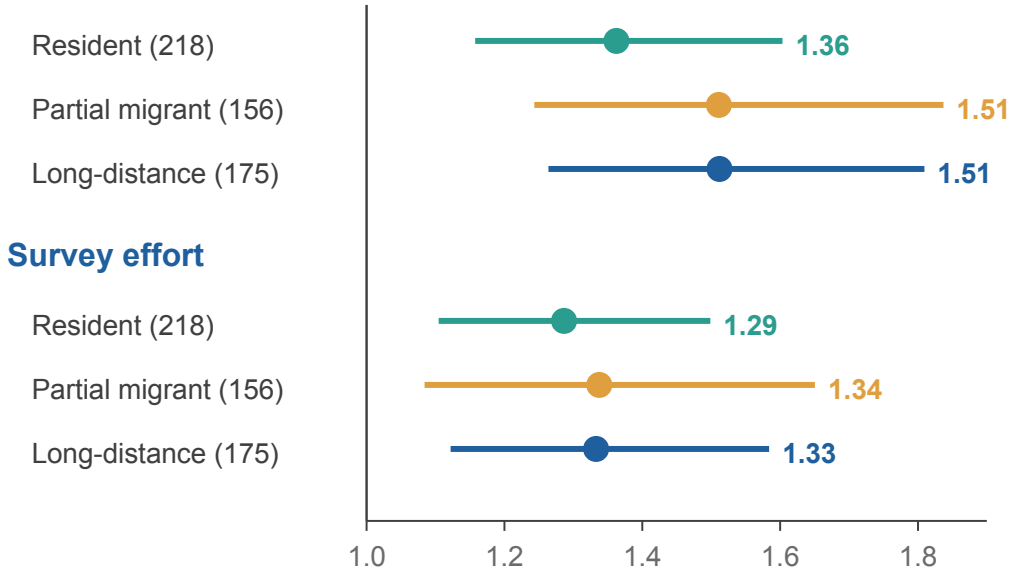
Partial migrants and wide-ranging species gain more records; drivers act in every group

Which species gain records? Odds ratio (95% CI)



Three lines per trait: taxonomic nesting, phylogenetic eigenvectors, phyloglm
1,298 species; reference: resident; continuous traits per SD; log scale

Both drivers, by migratory group: HR per SD

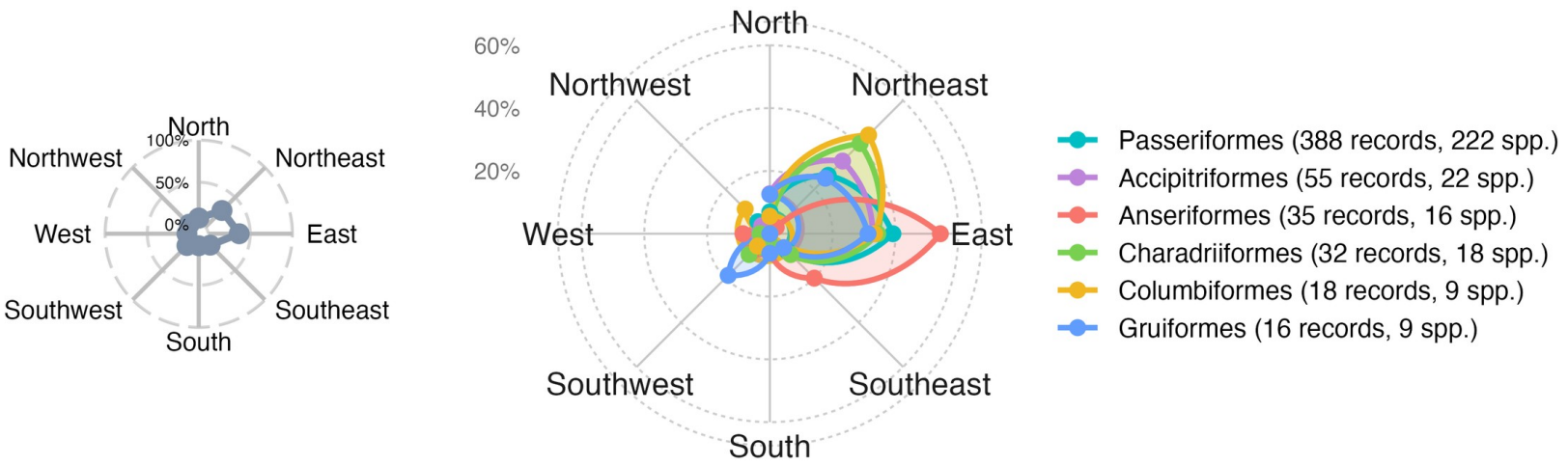


Events per group in brackets; groups fitted separately

Migratory strategy shifts the baseline (ΔAIC 6.2) but moderates neither driver (all LR P ≥ 0.26)

Finding 4

Seen from range centres, new records point east and north-east



Share of each order's own new records; absolute sample sizes are given in the legend

Bearings from range centroids: all records (left), six largest orders (right)

Most new records lie east or north-east of the species' range centre

Records

642

361 species, 20 orders

East

36.9%

237 of 642 records

North-east

27.9%

179 of 642 records

Finding 4

The eastward bias is geometry; net of it, records lean slightly north

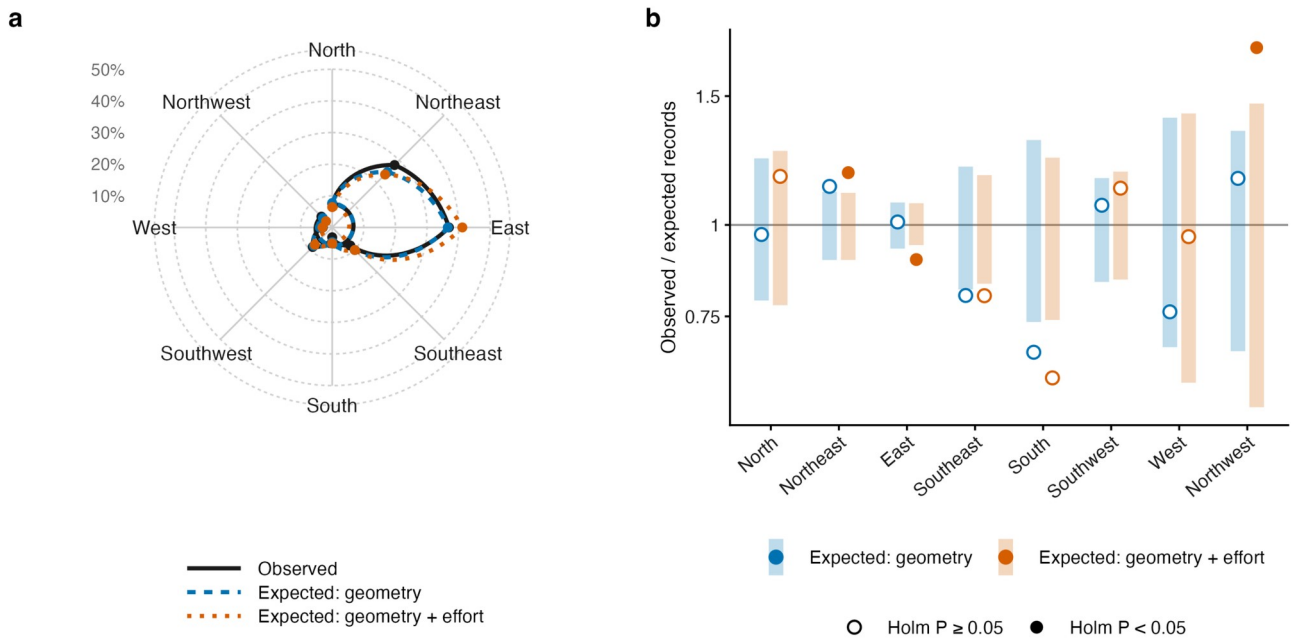


Fig. 6 Observed vs expected sectors under two nulls

- Null 1: each record placed at random among its species' reachable provinces
- Null 2: the same provinces, weighted by birding effort

Null 1: geometry only

East: 235 expected, 237 observed

No sector departs from the null

Null 2: geometry + birders

Northern sectors: 40.2% vs 32.8%

$\chi^2 = 26.8$, $P = 0.001$

NE, NW above; E below (Holm)

Net of geometry and effort: a modest poleward lean

Application

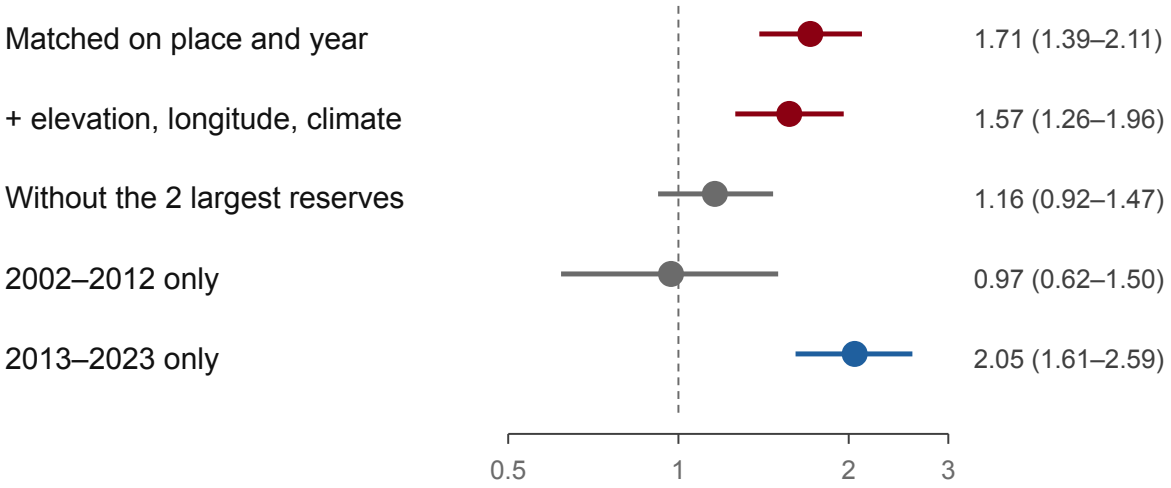
Records inside reserves track observation, not reserve performance

Share inside the 1,028 mapped reserves



Odds ratio against 200 matched controls

Controls from the same province and year, weighted by checklists; 95% CI



Few reserves carry the signal

172 events in 68 reserves

The top two hold 27.3% of them

Without them: OR 1.16

Only after birding boomed

2002–2012: OR 0.97

2013–2023: OR 2.05

24% more species per visit inside

Record counts should not index reserve performance

Limits

Four limits travel with these conclusions

Events are published records, not detections

We explain how records are documented, not range expansion itself.

Coordinates are precise to about 1 km at best

Reserve-overlap tests therefore use 0–5 km buffers as a sensitivity check.

Reserve polygons cover about a third of listed reserves

1,028 mapped reserves: no gap analysis, no 30×30 claims.

Projections leave the fitted support after 2030

Late-century maps are illustrations; the effort coefficient used is an upper bound.

Summary

Warming and survey effort leave signals of similar size

New records carry a climate signal mixed with an observation signal

01 Climate signal

Warming 1.36 vs effort 1.43 per SD: comparable, separable

02 Where few look

Warming × effort 0.84: under-surveyed regions hide the signal

03 Observation process

Province × year ×2.22; returns on effort 2.14 → 1.13

04 Who and where

Partial migrants gain more records; net of geometry, a slight north lean

Implications

- ☐ Send survey effort to climatically exposed, under-surveyed regions
- ☐ Read counts of new records as observation-weighted evidence, not as range shifts
- ☐ Test direction against geometric and effort nulls before calling it a shift

Birds or birders? Both. Model how we look, and new records tell us how ranges change

Thank you

Questions welcome

chenchen.ding@pku.edu.cn

Co-authors

- **Chenchen Ding**
  Institute of Ecology, College of Urban and Environmental Sciences, Peking University
- **Zhengwang Zhang**
  School of Life Sciences, Beijing Normal University
- **Zhigang Jiang**
  Institute of Zoology, Chinese Academy of Sciences
- **Zhiheng Wang**
  Institute of Ecology, College of Urban and Environmental Sciences, Peking University

Open data and code

CBNR v2 (CC BY 4.0) · doi.org/10.5281/zenodo.20809949

github.com/dingchenchen6/bird-newrecord-discovery-dated

Climate: WorldClim 2.1 / CRU TS 4.09 · Ranges: BirdLife International
Effort: eBird/GBIF and China Bird Report · Base map: GS(2019)1822

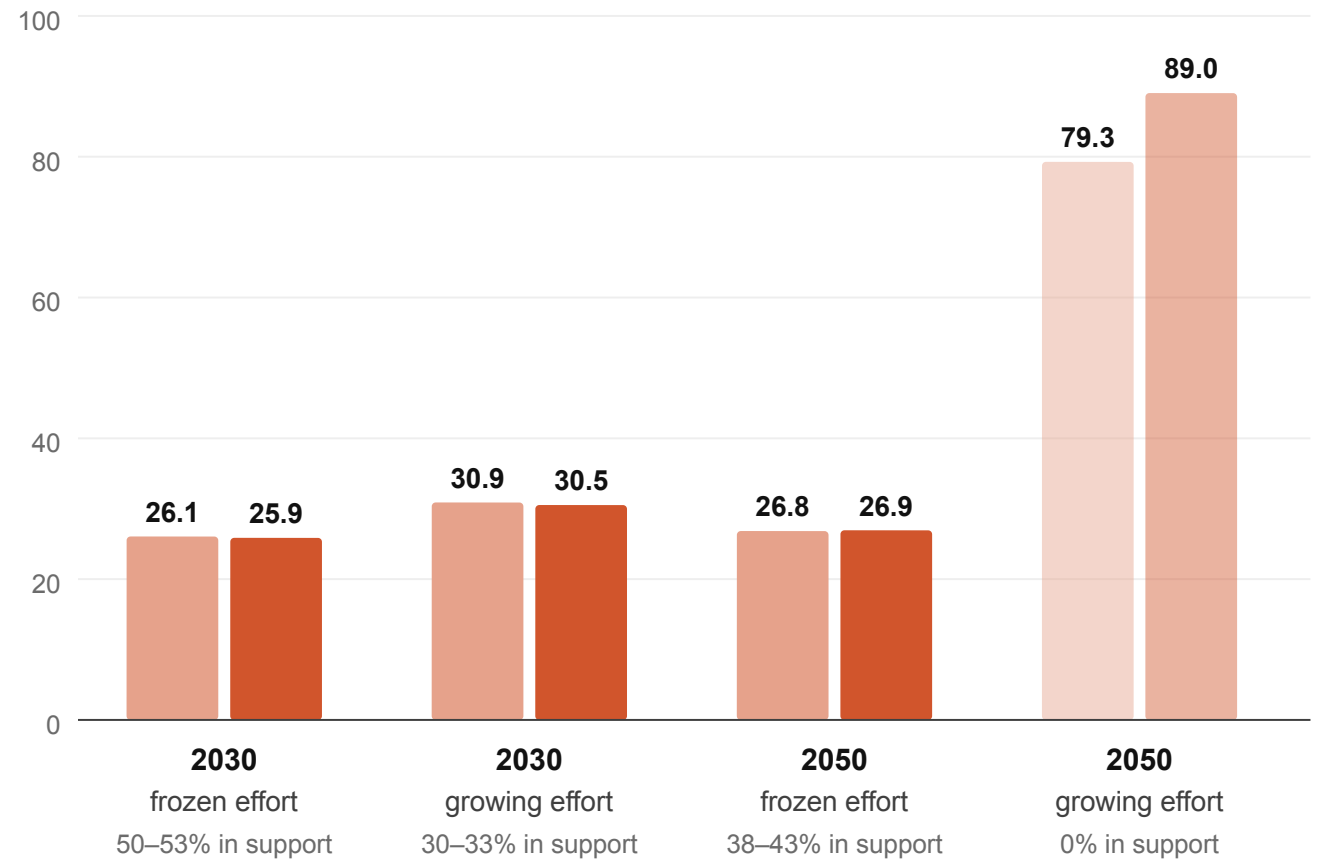
We thank birdwatchers across China, and every author whose records made CBNR possible.

CBNR data

Backup

Future record counts follow survey effort, not the emission scenario

Expected new records per year, China



Two climate × two effort scenarios

SSP2-4.5 or SSP5-8.5

Effort frozen, or still growing

What changes the count?

Climate scenario: barely anything

Growing effort: roughly triple

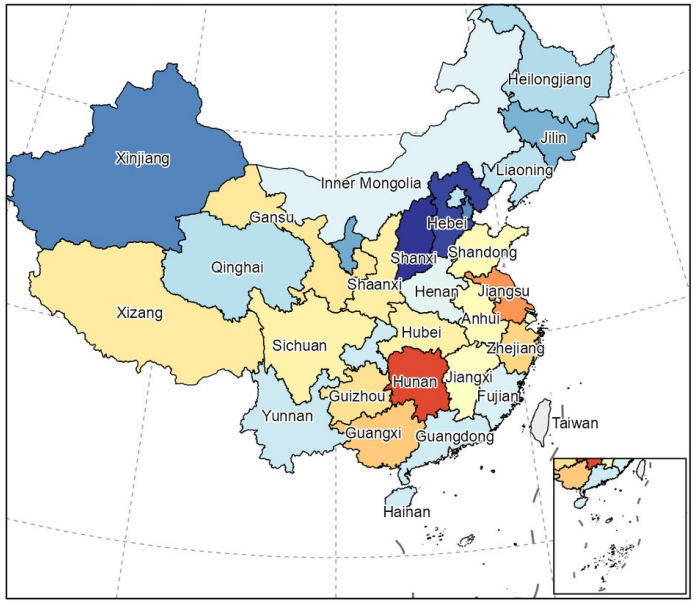
Most of that rise lies outside the fitted data (faded bars: none in support)

Backup

Records spread across central and southern China; growing effort leaves the data

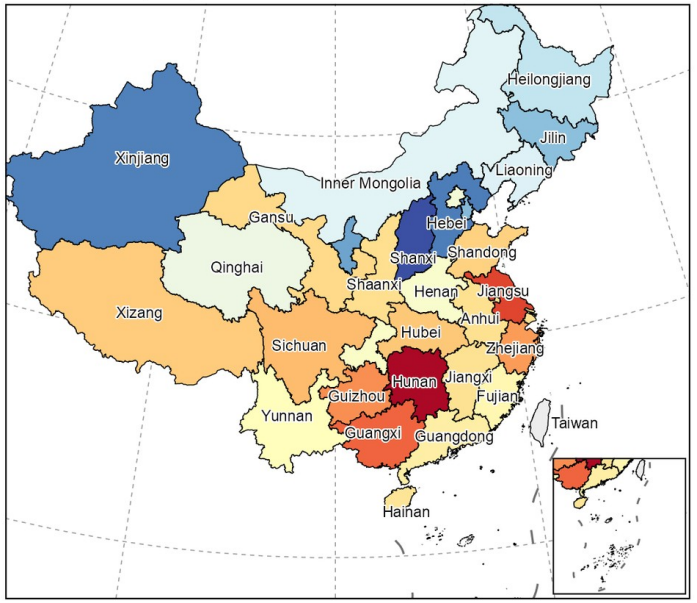
2030 · SSP2-4.5 · effort frozen at 2024

China 26.1 records per year · 53% of rows in support

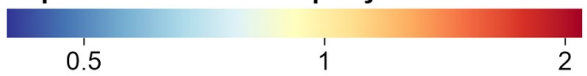


2030 · SSP2-4.5 · effort growing (+0.6 SD per decade)

China 30.9 records per year · 33% of rows in support



Expected new records per year



Percentages: share of species rows inside the fitted support. Grey: no effort data.
Under SSP5-8.5 the two effort scenarios give 25.9 and 30.5 records per year: almost the same.

2030 · SSP2-4.5 · expected new records per province; CBNR map standard, GS(2019)1822

Growing effort adds records in the south and east, beyond the data

Hunan, the top

1.7 → 2.1

per year: frozen → growing

Provinces off-data

8 → 16

< 50% of rows in support

Top five provinces

25%

of the national total

Backup

Data support, not ambition, sets the forecast horizon



Fig. 7b Hazard model vs machine learning

Support shrinks fast

Cells inside the fitted range:

2030: 46% → 2050: 6%

2080: none (SSP2-4.5)

Machine learning does not help

Agree inside the range (ρ = 0.56)

Diverge outside it

2030: usable
End of century:
illustration, not forecast

Backup

Dating to publication inflates effort and creates a phantom climate effect

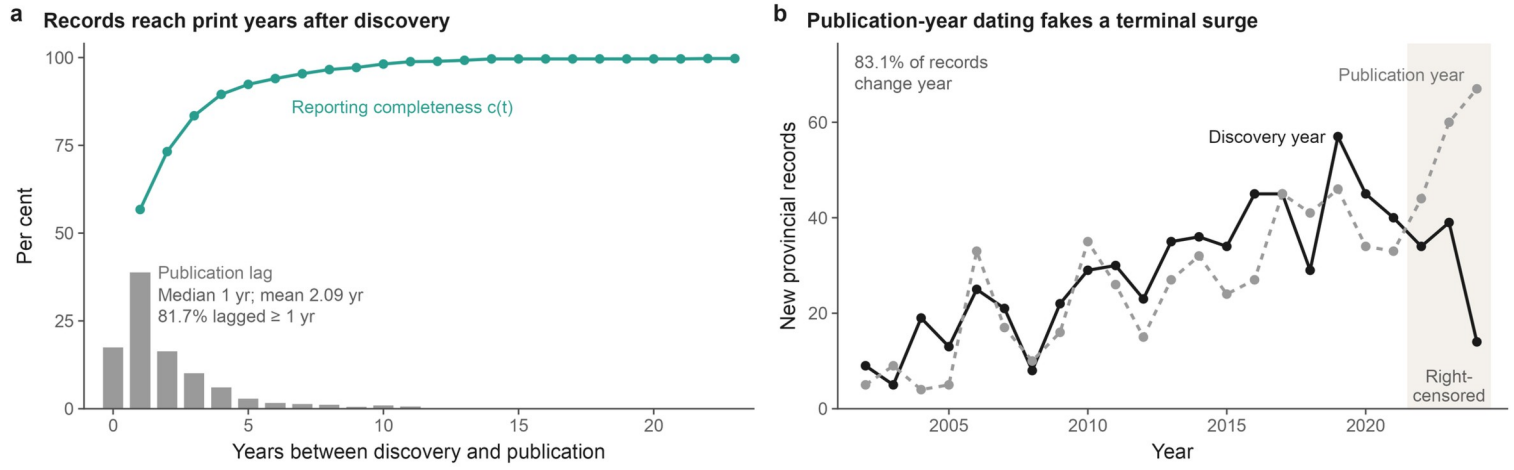


Fig. 2 Discovery vs publication dating

Backup

Specification robustness: climate indicator, window and effort proxy

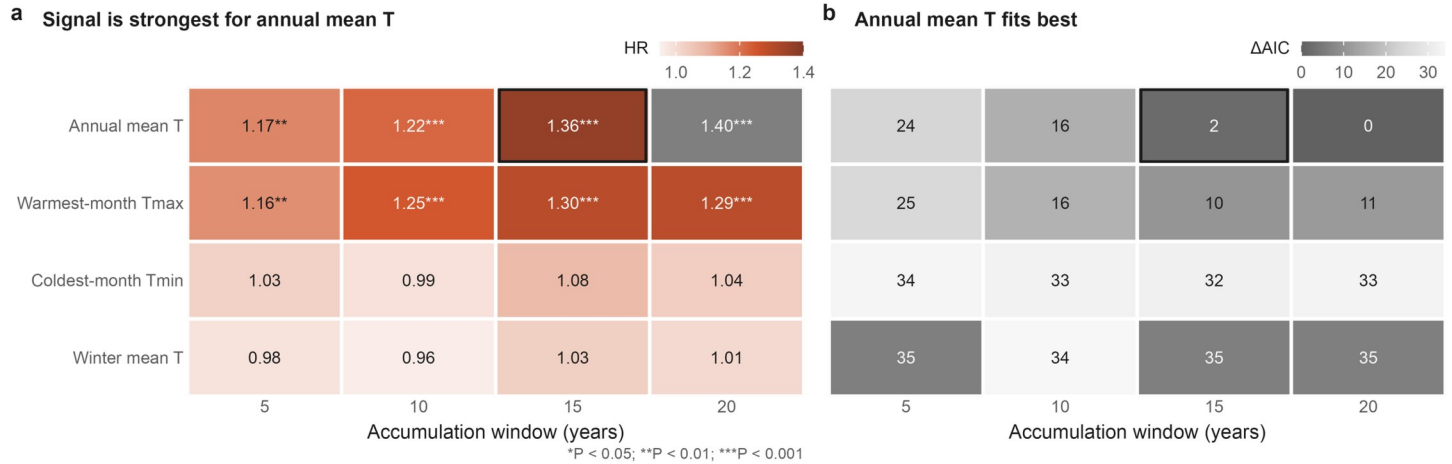


Fig. 3 Specification grid

Backup

Model anatomy: climate referenced to each species' range

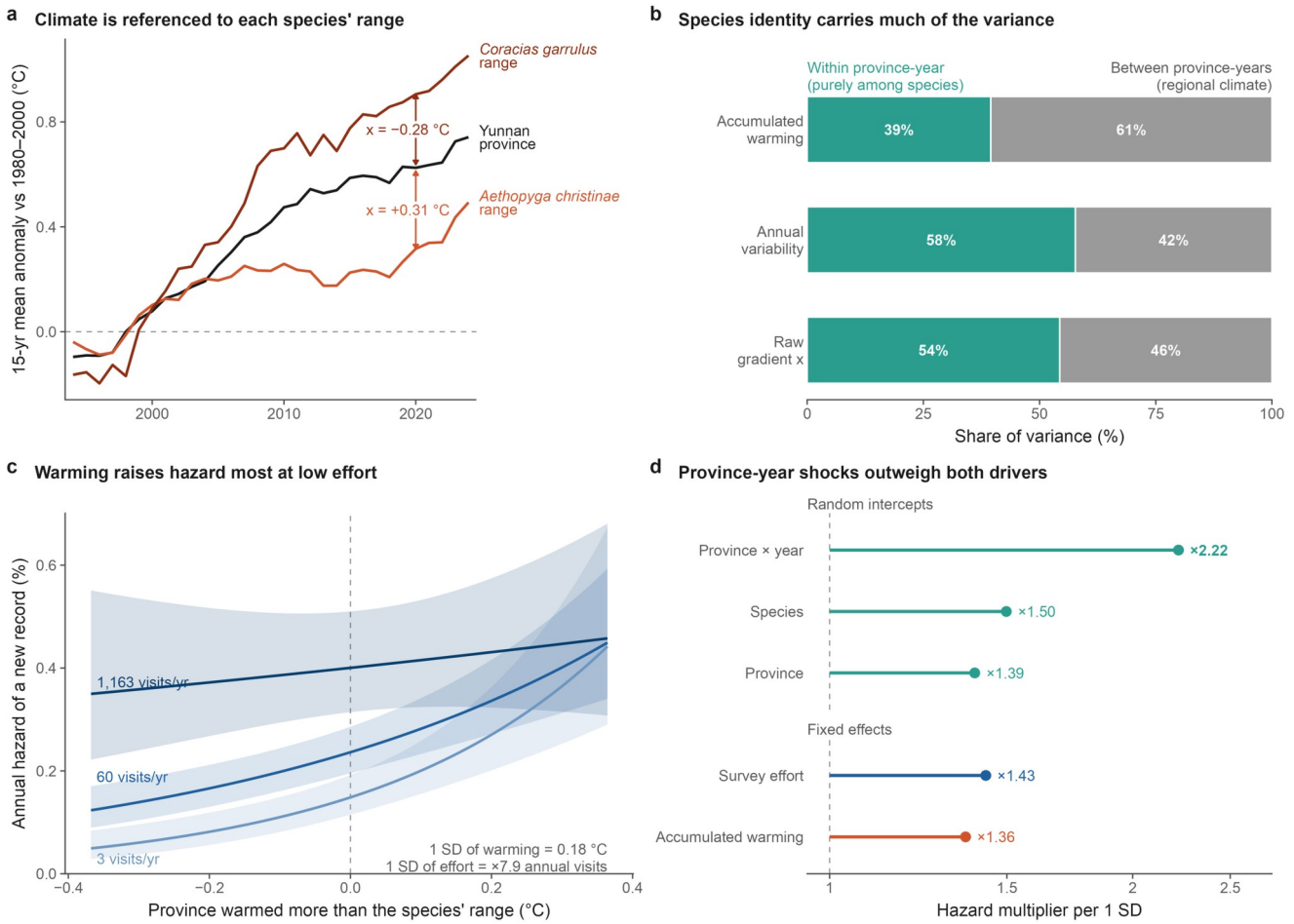
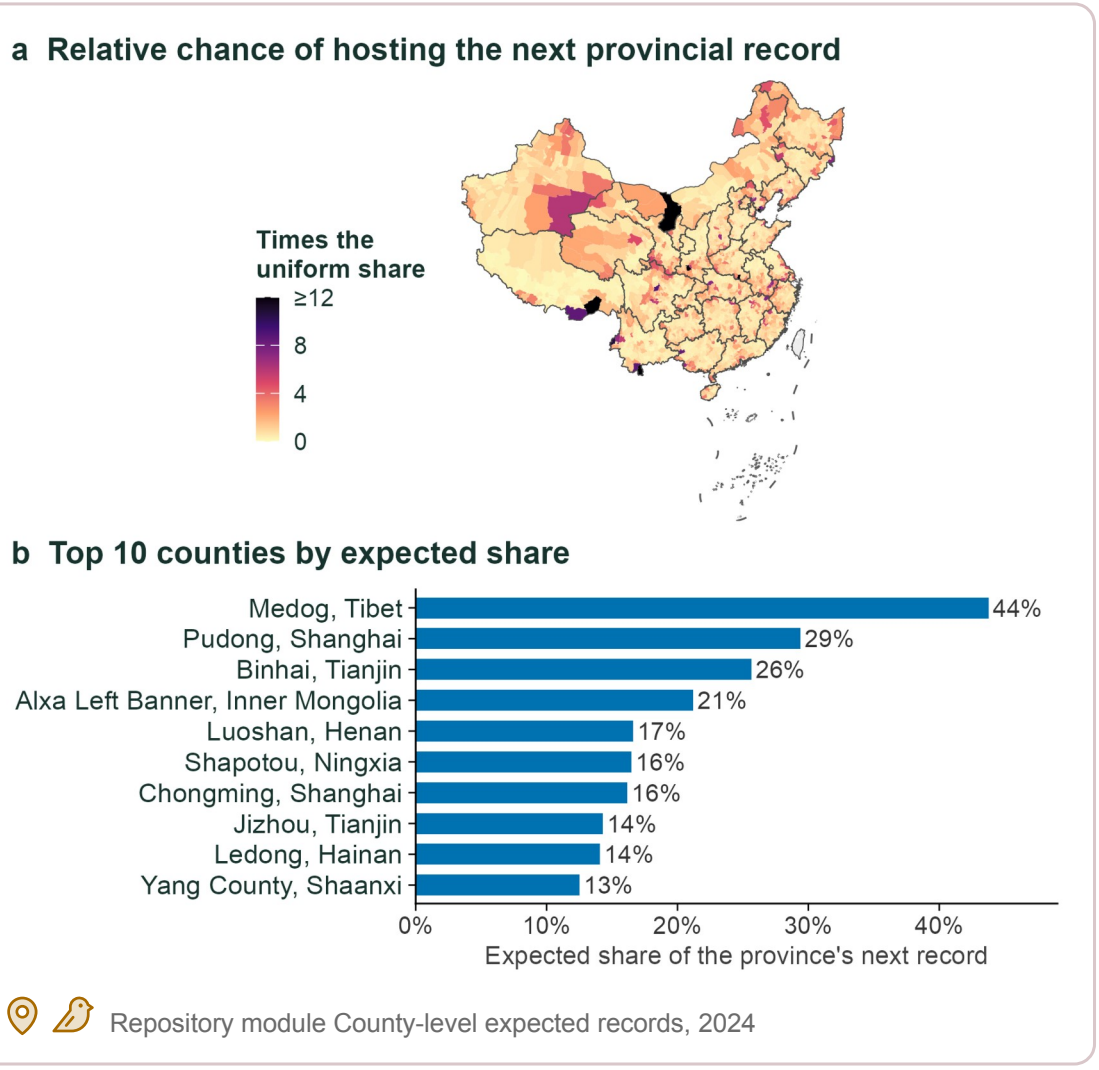


Fig. 4 Model anatomy

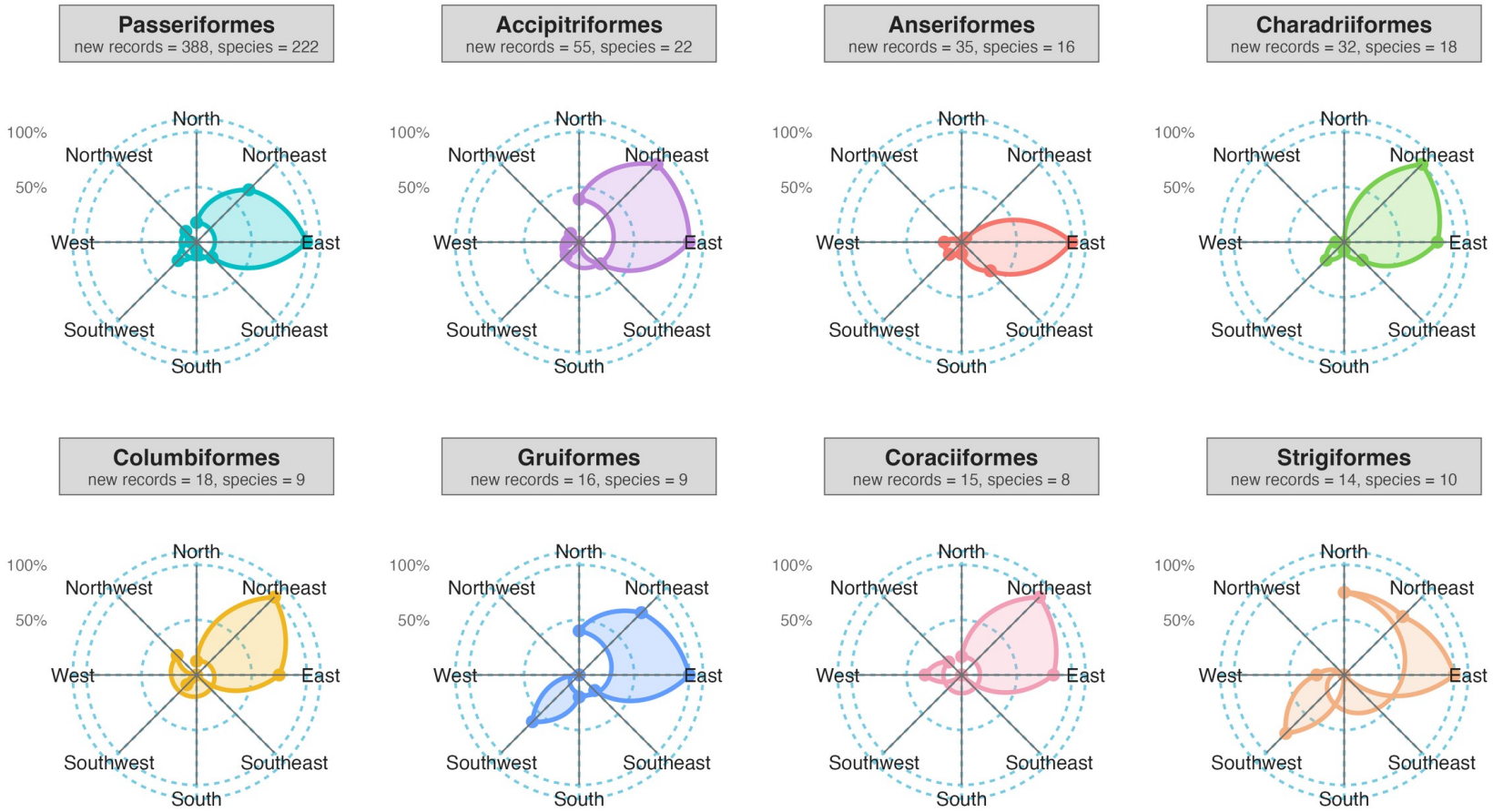
Backup

County-level surfaces: where to look first



Backup

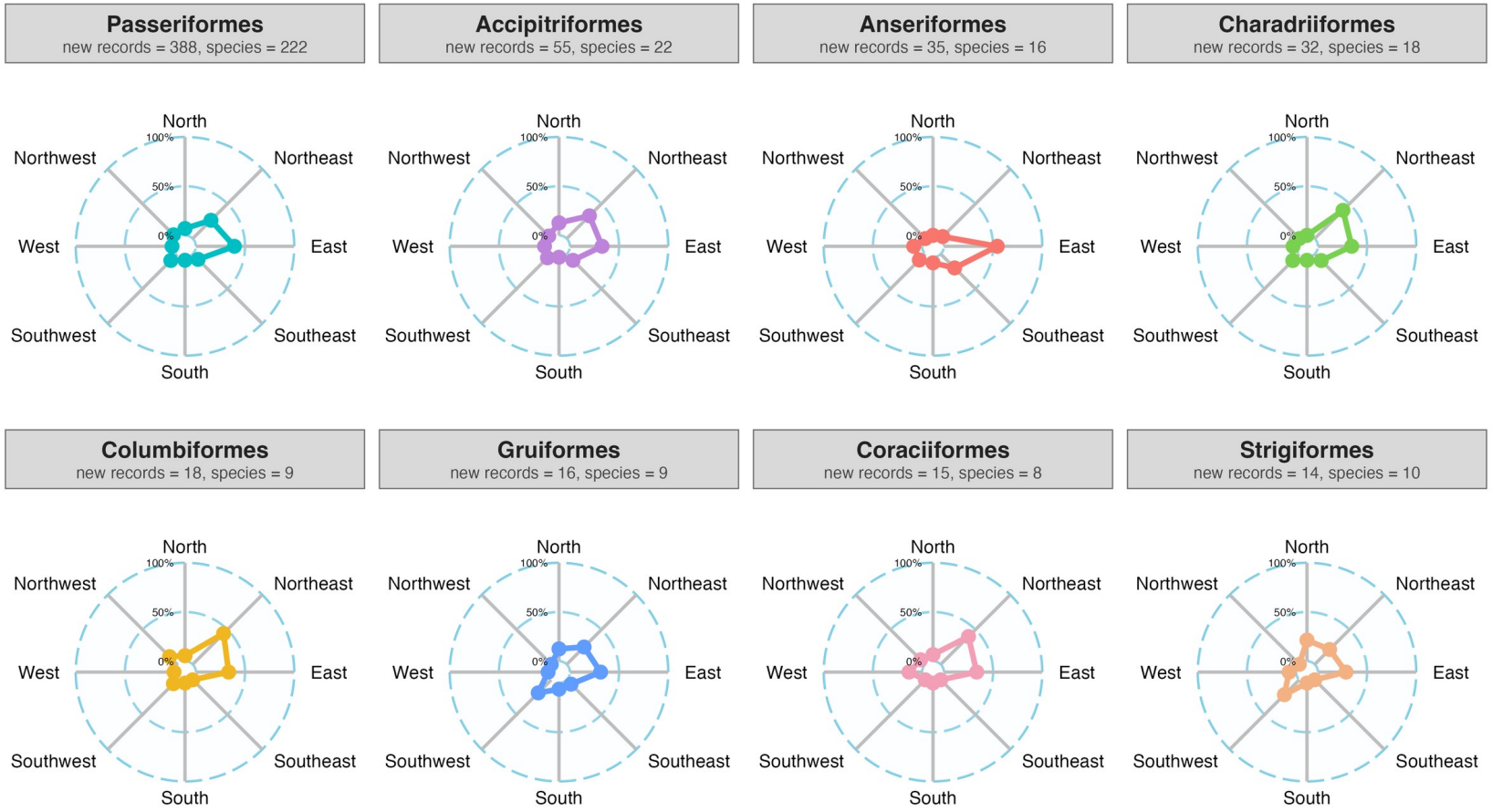
Direction by order: wind roses of the eight orders with most records



Share of each order's own records by compass sector

Backup

Direction by order: radar profiles of the eight orders with most records

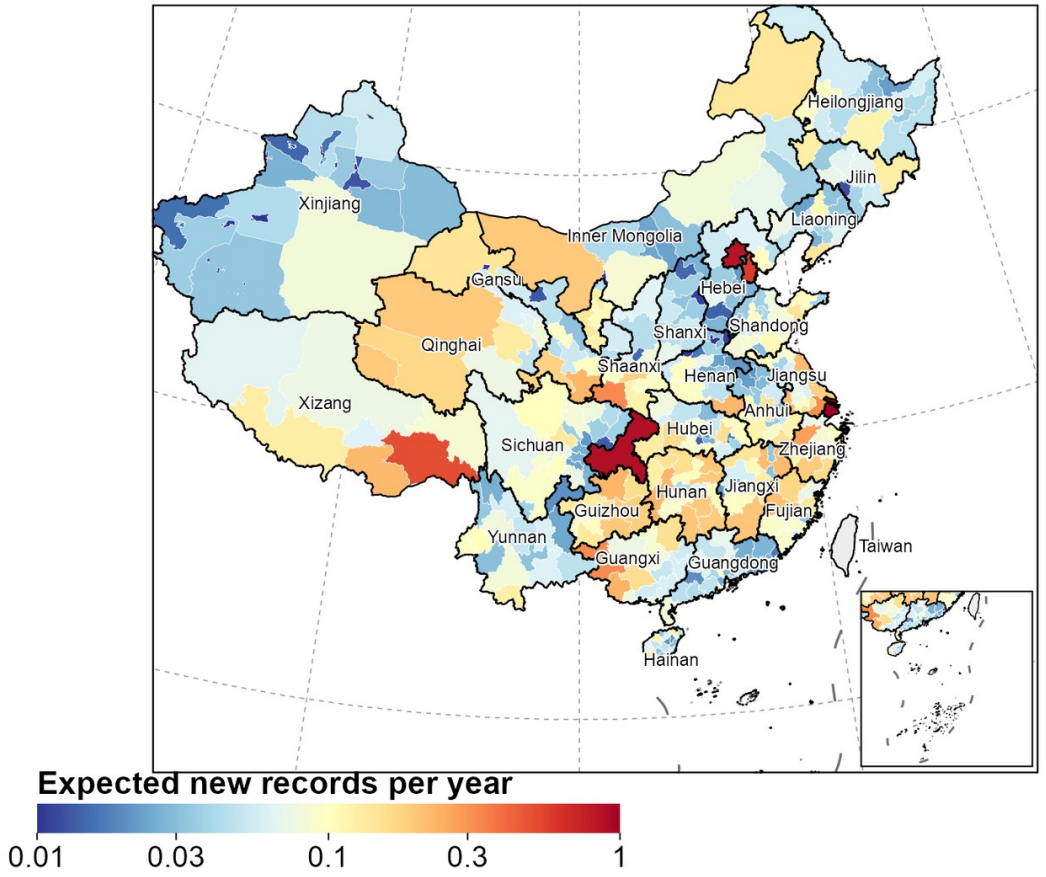


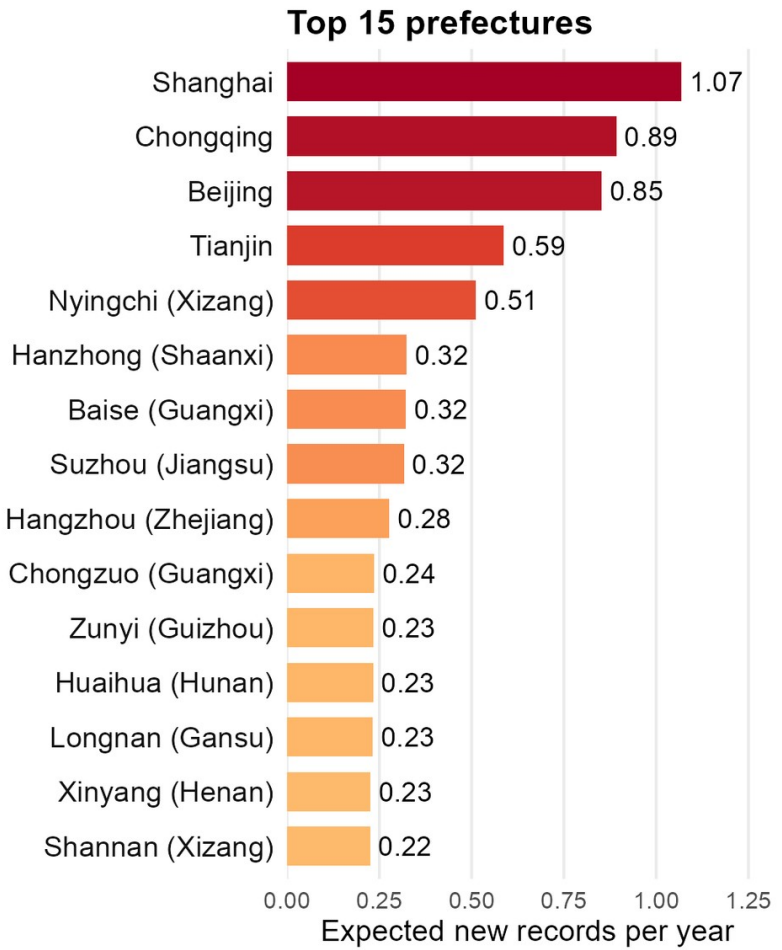
Radar profiles: share of each order's records by sector

Backup

Prefecture view of the 2030 futures: where to look first

2030 · SSP2-4.5 · effort growing: China 30.9 records per year

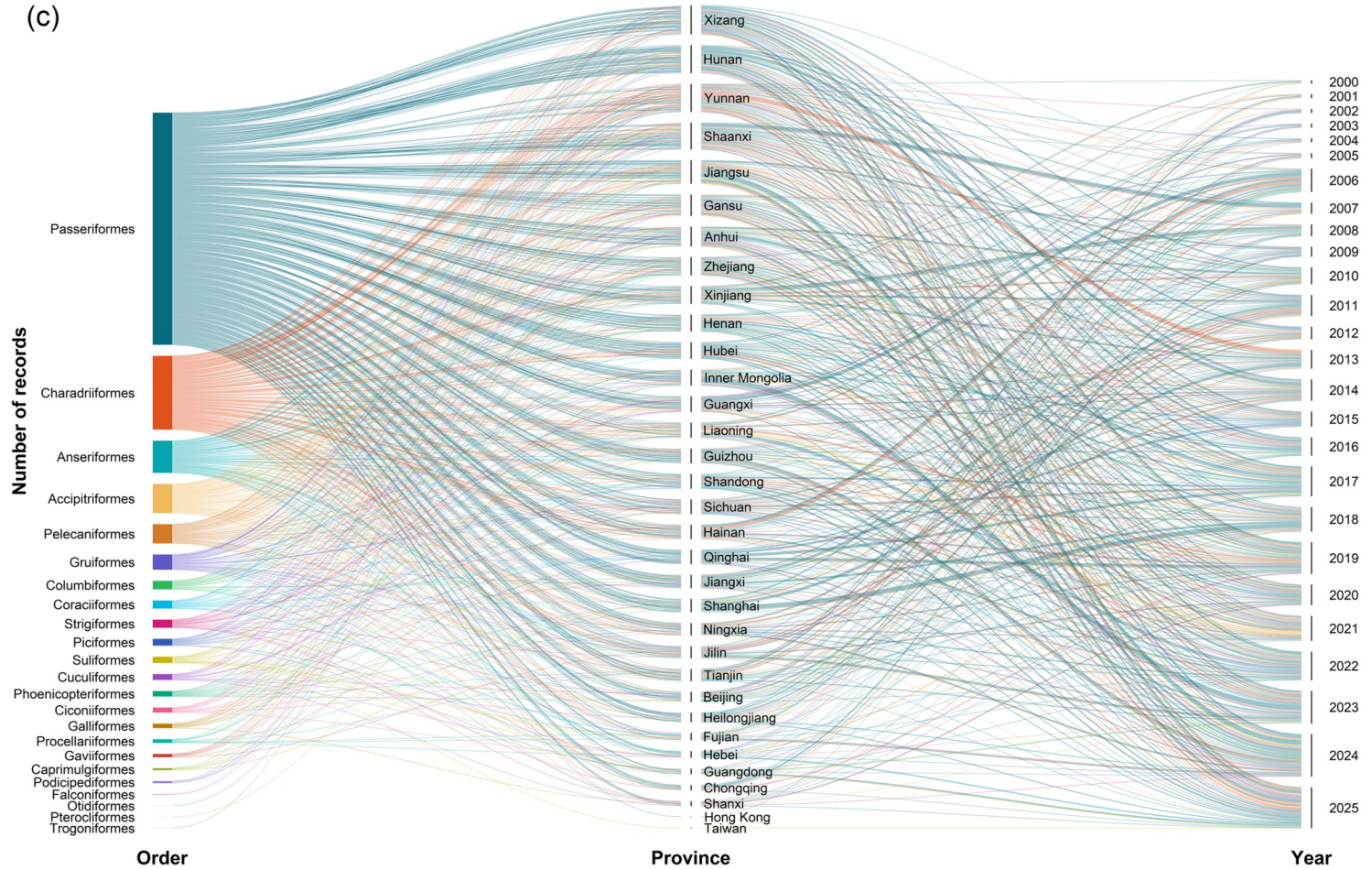




2030 · SSP2-4.5 · growing effort; prefecture shares held at 2024

Backup

CBNR Fig. 2c: records linked by order, province and publication year



CBNR Fig. 2c Ribbon width proportional to the number of records

Backup

Key references

Boakes, E. H. et al. (2010) PLoS Biol. 8, e1000385

Chen, I.-C. et al. (2011) Science 333, 1024–1026

Chen, S. et al. (2025) Avian Res. 16, 100310

Feng, X. et al. (2019) Bull. Chin. Acad. Sci. 34, 1389–1398

Herkt, K. M. B. et al. (2017) Glob. Ecol. Biogeogr. 26, 930–941

Hortal, J. et al. (2015) Annu. Rev. Ecol. Evol. Syst. 46, 523–549

Huang, G. et al. (2021) Natl. Sci. Rev. 8, nwab042

Hughes, A. C. et al. (2021) Ecography 44, 1259–1269

IPBES (2019) Global Assessment Report. IPBES Secretariat, Bonn

Jetz, W. et al. (2012) Nature 491, 444–448

Lahoz-Monfort, J. J. & Magrath, M. J. (2021) BioScience 71, 1038–1062

Lawlor, J. A. et al. (2024) Nat. Rev. Earth Environ. 5, 351–368

Lomolino, M. V. (2004) Frontiers of Biogeography, 293–296

Meyer, C. et al. (2015) Nat. Commun. 6, 8221

Mi, X. et al. (2021) Natl. Sci. Rev. 8, nwab032

Mittermeier, R. A. et al. (2011) Biodiversity Hotspots, 3–22

Oliver, R. Y. et al. (2021) PLoS Biol. 19, e3001336

Pacifici, M. et al. (2017) Nat. Clim. Change 7, 205–208

Pecl, G. T. et al. (2017) Science 355, eaai9214

Pimm, S. L. et al. (2014) Science 344, 1246752

Sanczuk, P. et al. (2026) Nat. Clim. Change 16, 21–25

Whittaker, R. J. et al. (2005) Divers. Distrib. 11, 3–23

Xing, X. et al. (2024) Front. Ecol. Evol. 12, 1415268

Zhang, R. (2011) China Animal Geography. Science Press

Zheng, G. (2023) Checklist of the Birds of China, 4th edn

Dataset: Ding et al., CBNR v2, Zenodo, doi.org/10.5281/zenodo.20809949

Q&A

Likely questions, with short answers (1 of 2)

Q1 **Is a single record a range shift, or just a vagrant?**

We never call one record a shift. We model the yearly chance of a first record, and warming and effort raise it in residents too, where vagrancy is rare.

Q2 **Isn't effort partly a response to records, with birders chasing rarities?**

Effort is all checklists per province-year, not trips for one species, and a province × year term absorbs pulses. Still, we read it as an association.

Q3 **How exactly do you measure warming?**

Accumulated warming in the province relative to the species' own range, in SD; strongest for annual mean temperature over 15–20 years (backups B2–B3).

Q4 **Why a discrete-time hazard model, not counts or an SDM?**

Each species–province pair is at risk every year until its first record; one model handles exposure, yearly climate and effort, and species and province effects.

Q5 **Why does dating to discovery matter so much?**

Publication lags discovery by about two years. Dated to publication, the effort effect inflates by about 37% and a phantom climate effect appears (backup B1).

Q&A

Likely questions, with short answers (2 of 2)

Q6 **How reliable is CBNR, if a language model read the papers?**

The model only speeds up reading: every record was checked by hand against its paper, plus taxonomy, coordinate and outlier checks. 1,076 records, open on Zenodo.

Q7 **What about land-use change and other drivers?**

Not in this model. Stable differences sit in the province term, year-to-year pulses in province × year. Land use is our next step.

Q8 **Records point east. Isn't that a range expansion?**

Geometry alone predicts it: 235 expected in the east, 237 observed. Net of geometry and effort, northern sectors hold 40% of records against 33% expected.

Q9 **What should monitoring programmes do with this?**

Warming × effort is 0.84: the climate signal is steepest where few people look. Send effort to under-surveyed regions that are warming fast.

Q10 **Can you predict where the next records will appear?**

Near term, yes: about 26–31 a year nationally in 2030, and the top-ranked county was right 16% of the time vs about 1% by chance. Beyond 2050 it is extrapolation.